

CSC3350 Software Development

Software Design Document

Group Team Project Spring 2026

Team 35

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Date: [mm/dd/yyyy]

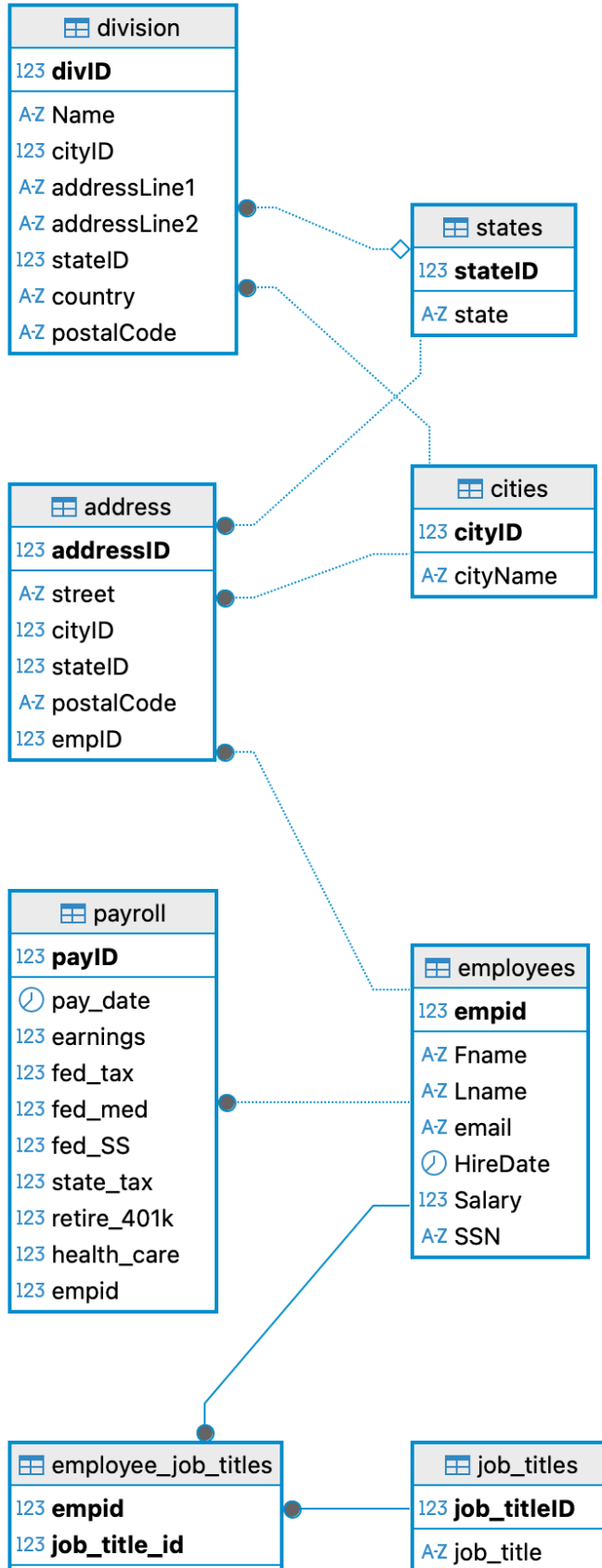
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1. INTRODUCTION

1.1. **Software Purpose**

[Identify the purpose of system being built and its intended audience.]

Document**2. DATABASE SCHEMA DIAGRAM**

3. JAVA CLASS DIAGRAMS

[Classes, interfaces, abstract classes showing details and relationships]

4. PROGRAMMING TASKS

PT-01 Update Employee Information

HR administrators may use this feature to update any information within an existing employee's profile. They will first select the employee they wish to edit by their unique identifier; next, they will see the current data of the employee's profile, and then they can enter new data in one or more fields (employee's first name, last name, e-mail, hire date, salary, SSN, or address ID) as needed. Once the administrator has entered the new information and the system has validated it and confirmed the existence of the employee, it will be saved to the database.

PT-02 Find an Employee to Delete

The system allows an HR Administrator to find an employee in the employee table to remove that employee's record. First, the administrator enters the employee's ID number. Then, the system searches the employee database for a matching record and displays the record to the administrator if found. The system then asks the administrator to confirm whether they want to delete the record. If they do, the system removes the record from the database. If there is no match, the system shows an error message and does nothing.

PT-03 Update Salaries By Percentage

This function of the system enables the HR Administrator to update employee salaries according to a certain percentage. The Administrator inputs a salary threshold and a percentage increase. The system searches through the employee database and finds all employees with

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salaries under the threshold. The system then raises those salaries by the provided percentage, and saves the changes to the database.

PT-04 — Employee Login Authentication

The system must allow users to log into the employee management system using a username and password. The system verifies the entered credentials against the employee database. If the credentials match a valid account, the user is granted access to the system. The system must also determine the user's role after login. HR administrators receive full permissions to manage employee records, update employee data, and generate reports. General employees are limited to viewing their own demographic and payroll information. If the username or password is incorrect, the system should deny access and display an error message indicating that the login attempt failed.

PT-05 Search Employee Data for Editing

The system allows an HR administrator to search for employee information using identifying details such as employee ID (empID), name, or SSN. Once the employee is found, the administrator can view the current employee record and select fields to edit. The system validates any updates and saves the changes to the database.

PT-06 Search Employee Data for Viewing

The system allows a general employee to search for employee information using identifying details such as name, date of birth (DOB), SSN, or employee ID (empID). The system searches the employee database and retrieves the matching employee record if it exists.

The system displays the employee's demographic and payroll information in view-only mode, meaning the user cannot modify or edit any of the information.

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If no matching employee record is found, the system should display a message indicating that no employee data was found.

PT-07 View Pay Statement History

The system allows a full-time employee to view their pay statement history after logging into the employee management system. The employee selects the option to view payroll history. The system retrieves the employee's payroll records from the database and displays the pay statements sorted by the most recent pay date first. The employee should only be able to view their own payroll records.

PT-08 Generate Reports

The system allows the HR administrator to generate employee payroll and hiring reports from the employee management system. After logging into the system, the HR administrator selects the report generation option and chooses one of the available report types. The system must support generating: total pay for a month by job title, total pay for a month by division, and new employee hires within a given date range. The system retrieves the necessary employee, payroll, division, and job title data from the database, processes the requested report, and displays the results in a readable format. If no matching data exists for the selected report criteria, the system should display a message indicating that no report data was found. This report functionality is limited to HR administrators only.

PT-09 Add Employee Divisions

The system allows an HR

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5. TEST CASES

Test Case ID: TC-01

Feature: Update Employee Information

Input: empid = 3, Field = email, New Value = lucy.new@example.com

Preconditions: An employee exists in the employee table who has the empid = 3.

Steps:

1. HR Admin logs into the system.
2. HR Admin chooses "Update Employee Information."
3. HR Admin inputs the employee id = 3.
4. The system displays the employee's present profile data (Lucy).
5. HR Admin chooses the email field and inputs lucy.new@example.com.
6. The system checks the new entry for validity and runs the "Update" SQL query.

Expected Outcome: The email field for the employee id = 3 is updated in the database. A "Select" query is run to verify the new entry.

Pass Conditions:

Updates the email field correctly → The new email value appears in the database immediately.

Fail Conditions:

Doesn't update the record → Old email remains after the action.

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Test Case ID: TC-02

Feature: Search Employee To Delete

Input: empid = 5

Preconditions: An employee with empid = 5 exists in the employee table.

Steps:

1. HR Admin logs into the system.
2. HR Admin chooses "Find an Employee to Delete."
3. HR Admin inputs the employee ID = 5.
4. The system searches the employee database and displays the matching employee's record.
5. The system asks the HR Admin to confirm deletion.
6. HR Admin chooses "yes" to confirm.

The system deletes the employee's record from the database.

Expected Result: The employee's record with empid = 5 is removed from the database. When the HR Admin searches again for empid = 5, the database says "no match."

Pass Conditions:

The system correctly identifies the record, has the administrator confirm, and removes the record from the database.

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Fail Conditions:

The system doesn't remove the record, the system can't find the record, the system won't ask for confirmation to delete the record, the system incorrectly removes another employee's record.

Test Case ID: TC-03

Feature: Update Salaries By Percentage

Input: salary threshold = 60000, percentage increase = 10%

Preconditions: There are employees in the database with salaries less than \$60,000.

Steps:

1. HR Admin logs into the system.
2. HR Admin selects "Update Salaries By Percentage."
3. HR Admin inputs salary threshold 60000.
4. HR Admin inputs percentage 10%
5. The system searches the employee database for employees with salaries lower than 60000.
6. The system increases those salaries.

Expected Result: All employees who earn less than \$60,000 have their salary increased by 10%.

Example:

Employee Salary: 50000 → 55000

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Pass Condition:

Salaries that fall under the threshold were raised correctly.

Fail Condition:

Salaries were not raised or were raised incorrectly.

Test Case ID: TC-04

Feature: Employee Login Authentication

Input:

username = valid_user

password = correct_password

Preconditions:

The employee account exists in the database.

Steps

1. User opens the employee management system.
2. User enters a username and password.
3. User selects the login button.
4. The system verifies the credentials in the database.

Expected Result:

The system successfully logs the user into the system and displays the main dashboard with permissions based on the user's role.

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Pass Condition:

User is granted access to the system and appropriate permissions are applied.

Fail Condition:

User is denied access even when valid credentials are entered.

Test Case ID: TC-05

Input: empID = 1000

Preconditions: Employee account with empID = 1000 exists first in the database; user is logged in as an HR Administrator.

Steps:

1. HR Admin logs into the employee management system.
2. HR Admin selects editing search employee data.
3. HR Admin enters the employee ID (1000).
4. The system searches the employee database and displays the matching employee record.
5. HR Admin reviews the current employee data.
6. HR Admin selects one or more fields to update (e.g., first name, email, hire date).
7. HR Admin enters new values for the selected fields.
8. The system validates the input and prompts for confirmation.
9. HR Admin confirms the changes.
10. The system updates the employee record in the database.

Expected Result:

The employee record for empID = 1000 is displayed and editable. After confirmation, the updated fields are correctly saved in the database.

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Pass Condition:

- Employee record is retrieved correctly.
- Selected fields are updated accurately.
- The database reflects the new values immediately after change.

Fail Condition:

- Employee record is not retrieved.
- The system allows editing invalid or unauthorized fields.
- Updates are not saved, or incorrect data is stored in the database.

Test Case ID: TC-06

Feature: Search Employee Data for Viewing

Input:

emplID = 1025

Preconditions:

The employee account exists in the database and the user is logged in as a general employee.

Steps

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1. User logs into the employee management system.
2. User selects Search Employee Data.
3. User enters employee ID 1025.
4. The system searches the employee database.
5. The system retrieves and displays the employee's information.

Expected Result:

The system displays the employee's demographic and payroll information in view-only mode.

Pass Condition:

Employee data is displayed correctly and cannot be edited.

Fail Condition:

Employee data is not displayed or the system allows editing.

Test Case ID: TC-08

Feature: Generate Reports

Input:

- report type = Total Pay by Division
- month = January 2026

Preconditions:

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HR Administrator is logged into the system and payroll/division data exists in the database for January 2026.

Steps

1. HR Admin logs into the employee management system.
2. HR Admin selects Generate Reports.
3. HR Admin selects the report type Total Pay by Division.
4. HR Admin enters the month January 2026.
5. The system retrieves payroll records and related division data from the database.
6. The system calculates the total pay for each division for the selected month.
7. The system displays the completed report.

Expected Result:

The system displays a report showing each division and the total employee pay for January 2026.

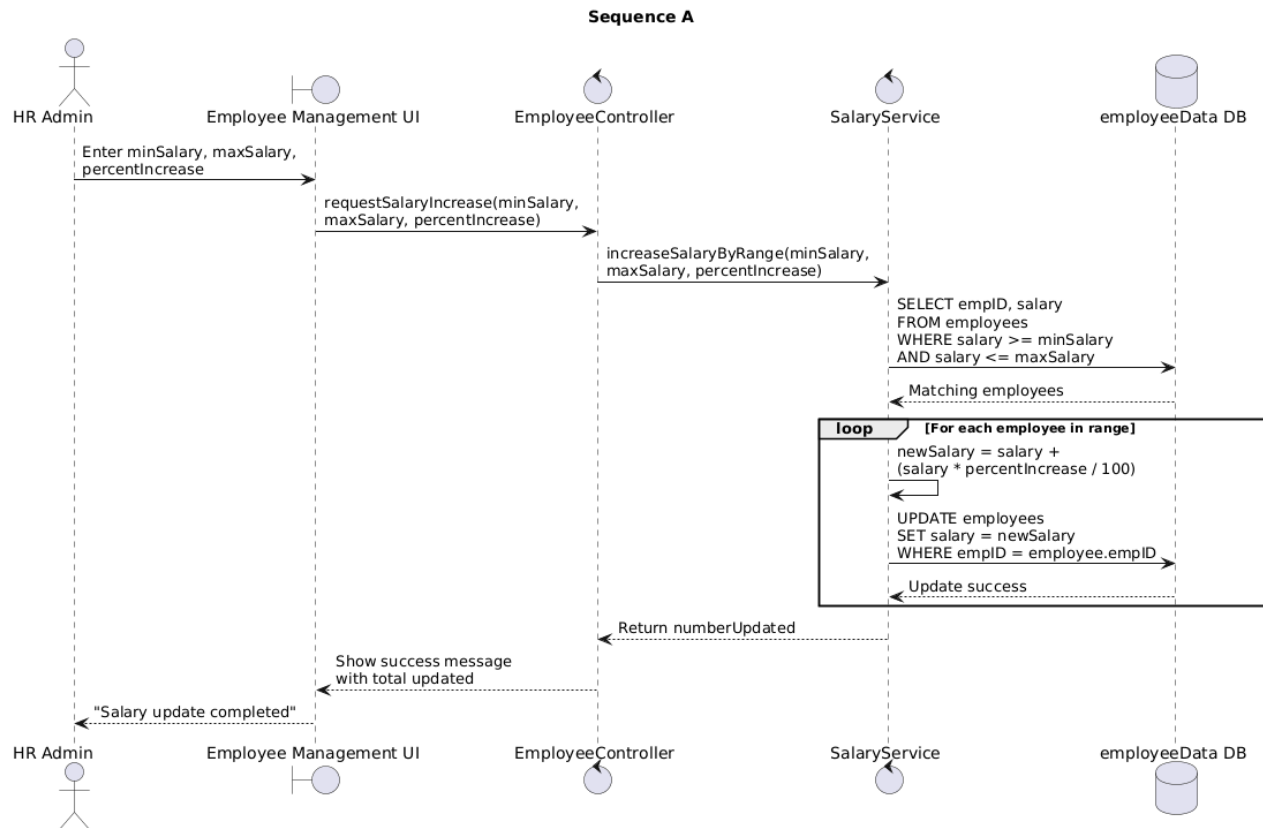
Pass Condition:

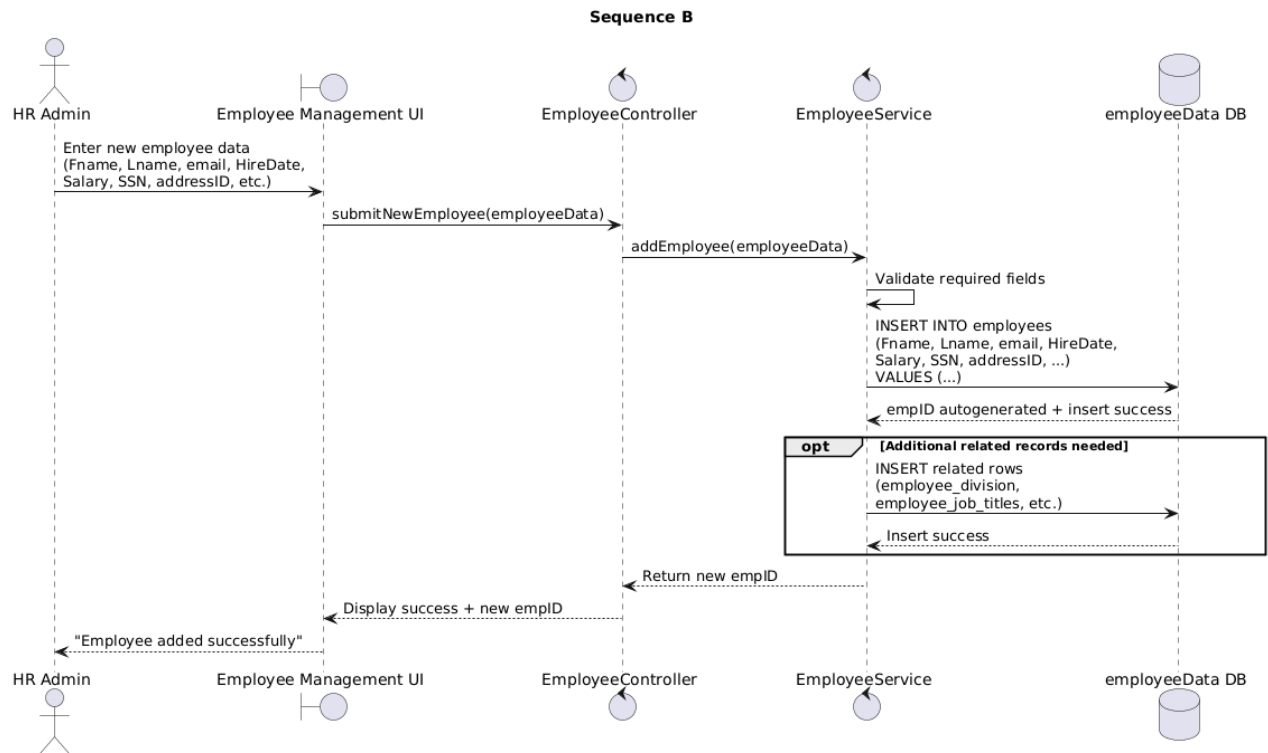
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The correct report is generated, the totals are calculated accurately, and the report is displayed only to the HR Administrator.

Fail Condition:

The report is not generated, totals are incorrect, division data is missing or mismatched, or access is allowed for a general employee.

Document**6. SEQUENCE DIAGRAMS**

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7. APPENDIX

7.1. Definitions and Acronyms

7.2. Additional UML diagrams

7.3. Reflection

[give a short paragraph of your impression/opinion of the project: importance, understandability, instruction quality, clarity of directions, timeline.]